THE ANSWERS

Notations:

(a) B - Bookwork

(b) E - New example

(c) A - New application

1. a) i) $P\left(X^2 \geq 2a | X^2 > a\right) = \frac{P\left(X^2 \geq 2a\right)}{P(X^2 > a)}$. [1 - A]

We can then compute $P\left(X^2 > a\right) = P\left(X > \sqrt{a}\right) = e^{-a}$. [2 - A]

We can then conclude that $P\left(X^2 \geq 2a | X^2 > a\right) = \frac{e^{-2a}}{e^{-a}} = e^{-a}$. [1 - A]

ii) $E(X) = \int_0^\infty x 2x e^{-x^2} dx = \int_{-\infty}^\infty x^2 e^{-x^2} dx$. [1 - A]

Writing $x = \frac{u}{\sqrt{2}}$, we find

$$E(X) = \frac{1}{2\sqrt{2}} \int_{-\infty}^{\infty} u^2 e^{-u^2/2} du = \frac{\sqrt{2\pi}}{2\sqrt{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} u^2 e^{-u^2/2} du.$$

[2 - A]

Hence, $E(X) = \frac{\sqrt{\pi}}{2} E(Z^2) = \frac{\sqrt{\pi}}{2}$, with $Z \sim N(0,1)$. [1 - A]

iii) Since $P\left(X^2 > a\right) = e^{-a}$, it is clear that X^2 is an exponentially distributed random variable with parameter $\lambda = 1$, i.e. $X^2 \sim EXPO(1)$. [3 - A]

iv) Since $E(X^2) = 1$, $\text{Var}(X) = 1 - E(X)^2 = 1 - \frac{\pi}{4}$. [3 - A]

b) i) Recall

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

The likelihood function is then

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)} = \frac{1}{(2\pi)^{n/2}(\sigma^2)^{n/2}} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^n (x_i-\mu)^2}$$

[2 - B]

The log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

We seek the values of μ and σ^2 that jointly maximise the log-likelihood. Consider μ first:

$$\frac{\partial \ell(\mu, \sigma^2)}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu).$$

Equating with zero yields

$$\frac{1}{\hat{\sigma}^2}\sum_{i=1}^{n}(x_i-\hat{\mu})=0 \quad \Rightarrow \quad \sum_{i=1}^{n}x_i-n\hat{\mu}=0\,,$$

and the MLE is

$$\hat{\mu}=\frac{1}{n}\sum_{i=1}^{n}X_i\,.$$

[2 - B]

For σ^2 we have

$$\frac{\partial \ell(\mu,\sigma^2)}{\partial \sigma^2}=-\frac{n}{2}\frac{1}{\sigma^2}+\frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i-\mu)^2\,,$$

which we equate with zero to obtain

$$-\frac{n}{2}\frac{1}{\hat{\sigma}^2}+\frac{1}{2\hat{\sigma}^4}\sum_{i=1}^{n}(X_i-\hat{\mu})^2=0\Rightarrow\hat{\sigma}^2=\frac{1}{n}\sum_{i=1}^{n}(X_i-\bar{X})^2\,.$$

[2 - B]

ii) We check the second order derivatives and make sure they are negative.

$$\frac{\partial^2 \ell}{\partial \mu^2}=\frac{-1}{\sigma^2}\leq 0.$$

[2 - A]

$$\frac{\partial^2 \ell}{\partial (\sigma^2)^2}=\frac{n}{2}\frac{1}{\sigma^4}-\frac{1}{\sigma^6}\sum_{i}(x_i-\mu)^2.$$

Evaluating at $\hat{\mu},\hat{\sigma}^2$, we need to show that $\frac{\partial^2 \ell}{\partial (\sigma^2)^2}\Big|_{\hat{\mu},\hat{\sigma}^2}\leq 0$. [1 - A]

We get

$$\frac{\partial^2 \ell}{\partial (\sigma^2)^2}\Big|_{\hat{\mu},\hat{\sigma}^2}=\frac{n}{2}\frac{1}{\hat{\sigma}^4}-\frac{1}{\hat{\sigma}^6}\sum_{i}(x_i-\hat{\mu})^2$$

which is re-written using $n\hat{\sigma}^2=\sum_{i=1}^{n}(x_i-\hat{\mu})^2$ as

$$\frac{\partial^2 \ell}{\partial (\sigma^2)^2}\Big|_{\hat{\mu},\hat{\sigma}^2}=\frac{n}{2}\frac{1}{\hat{\sigma}^4}-\frac{n}{\hat{\sigma}^4}=-\frac{n}{2}\frac{1}{\hat{\sigma}^4}\leq 0.$$

[2 - A]

2. a) i) We first write $\text{Cov}(Y_1,Y_2) = E(Y_1Y_2) - E(Y_1)E(Y_2)$.

$$E(Y_1Y_2) = \int_0^1 \int_0^1 y_1 y_2 2y_1 dy_1 dy_2 = 1/3.$$

[1 - E]

$$E(Y_1) = \int_0^1 \int_0^1 y_1 2y_1 dy_1 dy_2 = 2/3.$$

$$E(Y_2) = \int_0^1 \int_0^1 y_2 2y_1 dy_1 dy_2 = 1/2.$$

[1 - E]

Then, $\text{Cov}(Y_1,Y_2) = 0$ and $\text{Corr}(Y_1,Y_2) = 0$. [1 - E]

ii) We first write

$$P(Y_1 \leq 0.25 | Y_2 \geq 1/3) = \frac{P(Y_1 \leq 0.25 \bigcap Y_2 \geq 1/3)}{P(Y_2 \geq 1/3)}.$$

[1 - E]

The numerator is given by

$$\int_0^{0.25} \int_{1/3}^1 2y_1 dy_2 dy_1 = \frac{2}{3}\frac{1}{16}.$$

[1 - E]

The denominator is given by

$$\int_0^1 \int_{1/3}^1 2y_1 dy_2 dy_1 = \frac{2}{3}.$$

[1 - E]

Hence, $P(Y_1 \leq 0.25 | Y_2 \geq 1/3) = \frac{1}{16}$.

iii) $\text{Var}(3Y_1 - 2Y_2 + 5) = E((3(Y_1 - \mu_{Y_1}) - 2(Y_2 - \mu_{Y_2}))^2)$. Since $\text{Cov}(Y_1,Y_2) = 0$, $\text{Var}(3Y_1 - 2Y_2 + 5) = 9\text{Var}(\text{Y}_1) + 4\text{Var}(\text{Y}_2)$. [2 - E]
We compute $\text{Var}(Y_1) = 1/18$ [1 - E]
We compute $\text{Var}(Y_2) = 1/12$ [1 - E]
Finally, $\text{Var}(3Y_1 - 2Y_2 + 5) = 5/6$.

iv) They are uncorrelated since $\text{Cov}(Y_1,Y_2) = 0$. [1 - E]
They are independent since $f_{Y_1,Y_2}(y_1,y_2) = f_{Y_1}(y_1) f_{Y_2}(y_2)$, $\forall y_1, y_2$. Indeed, $f_{Y_1}(y_1) = 2y_1$ for $0 < y_1 < 1$ and 0 otherwise, and $f_{Y_2}(y_2) = 1$ for $0 < y_2 < 1$ and 0 otherwise. [2 - E]

b) i) Since X and Y are exponentially distributed and independent,

$$f_{X,Y}(x,y) = \begin{cases} e^{-(x+y)}, & x > 0, y > 0 \\ 0, & \textit{otherwise}. \end{cases}$$

[4 - A]

ii) We make the change of random variable $U = XY$ and $V = Y$, i.e. $Y = V$ and $X = U/V$. We then compute $\det(J) = 1/v$ [2 - A]
and

$$f_{U,V}(u,v) = \begin{cases} \frac{1}{v} e^{-(\frac{u}{v} + v)}, & u > 0, v > 0 \\ 0, & \textit{otherwise}. \end{cases}$$

[3 - A]

Hence,

$$P(U \leq \varepsilon) = \int_0^{\varepsilon} \int_0^{\infty} \frac{1}{v} e^{-(\frac{u+v^2}{v})} dv du.$$

This gives

$$P(U \leq \varepsilon) = \int_0^{\infty} (1 - e^{-\varepsilon/v}) e^{-v} dv = 1 - \int_0^{\infty} e^{-(\varepsilon/v+v)} dv = 1 - 2\sqrt{\varepsilon} K_1\left(2\sqrt{\varepsilon}\right)$$

. [3 - A]